



OPEN Combating cyberbullying with transparency: unveiling the impact of IP location disclosure on cyberbullying in Chinese social media

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The Internet has transformed into a vital communication hub in daily life, yet cyberbullying has concurrently escalated, posing a critical societal concern. To tackle this issue, the Chinese government has mandated that major social media platforms disclose users' IP locations. However, it is currently unclear about the effectiveness of this measure in reducing cyberbullying and the underlying mechanisms. Drawing on impression management theory, we conducted three experiments across three prominent social platforms in China, aiming to explore how the display of IP locations affects cyberbullying through impression management motivation and the contingent role of group identity. Our findings indicate that disclosing IP locations decreases cyberbullying behavior by bolstering individuals' impression management motivation. Notably, this effect is more evident among those with a stronger sense of group identity. Moreover, we rigorously examined and excluded alternative explanations, such as moral disengagement and privacy concerns, ensuring the robustness of our findings. This research not only offers theoretical enlightenment on suppressing cyberbullying from an IT design standpoint but also extends practical guidance to internet policymakers, social platform managers, and all other stakeholders implicated in the issue of cyberbullying.

Keywords IP display, Cyberbullying, Impression management motivation, Group identity

With the rapid development of internet technology, cyberspace has become an indispensable platform for people's daily communication, information acquisition, and sharing. However, while internet anonymity protects personal privacy, it also provides fertile ground for the breeding of cyberbullying. Cyberbullying refers to the threatening, aggressive, and insulting behavior carried out on other people or groups through text, pictures, and videos on the internet¹. Reports show that over 40% of adult Internet users have been victims of cyberbullying, and 75% have witnessed cyberbullying in their surroundings (<https://www.broadbandsearch.net/blog/cyber-bullying-statistics#post-navigation-1>, accessed 25 August 2024). Cyberbullying not only poses a severe threat to individuals' physical and mental health, but also challenges the harmony of the online environment and social stability^{2,3}. Therefore, it is necessary to pay attention to the phenomenon of cyberbullying and explore how to prevent it effectively.

Previous studies have proposed several measures to prevent cyberbullying, including anti-cyberbullying education⁴, algorithm monitoring^{5–7}, bystander intervention^{5,8}, and so on. However, the applicability and effectiveness of these measures are often less than satisfactory. Specifically, anti-cyberbullying education usually requires significant investment and can only reach a limited audience, algorithm monitoring relies on the continuous improvement of models and often lags behind the evolution of cyberbullying, and bystander intervention cannot address the root cause of cyberbullying and has relatively low controllability. In contrast, the Chinese government, starting from August 2022, has implemented a mandatory policy requiring all major social platforms to display users' IP locations. This intervention measure aims to regulate users' online conduct and purify the internet environment by reducing user anonymity. It boasts advantages such as being

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tailored to platform design, addressing the root of cyberbullying, offering low cost, extensive reach, and high controllability. Yet, there is a dearth of empirical evidence to substantiate whether disclosing users' IP locations can effectively alleviate the instances of cyberbullying they perpetrate. Exploring how IP display specifically impacts cyberbullying behaviors not only aids in deepening our understanding of the causes and evolution of cyberbullying but also offers theoretical foundations and practical guidance for formulating more scientific and reasonable internet regulation policies. Toward this end, our research objectives are: (1) to investigate the impact of IP location display on cyberbullying, (2) to reveal the underlying mechanisms from the perspective of user psychology, and (3) to explore individual characteristics as contingent factors between the relationship of IP location display and cyberbullying.

To answer the aforementioned questions, we first need to clarify what IP location display means to users. As a part of personal real information, IP location display reduces the perceived anonymity of information senders when facing others⁹, and implies a more easily identifiable user identity¹⁰. Under the circumstances, individuals may feel greater scrutiny and evaluation from others, which affects the self-image they present externally^{10–12}. Therefore, we introduce impression management theory to explain the mechanism by which IP location display affects cyberbullying¹³. In particular, we view an individual's behavior to reduce participation in cyberbullying as a defensive impression construction to avoid damage to their image. This behavior is activated by impression management motivation, and IP location display can be considered as an environmental cue that triggers an individual's impression management motivation. Additionally, within the social networking environment, individuals often belong to specific social groups, and this sense of belonging forms group identity¹⁴. Group identity plays a significant role in shaping individuals' attitudes and behaviors¹⁵. In the case of IP location display, the sense of identity with the geographical group to which they belong will affect individuals' motivation and behavior to maintain the group's image. Therefore, we incorporate users' group identity to explore its potential moderating role between IP location display, impression management motivation, and cyberbullying behavior, in order to gain a deeper understanding of the complex social dynamics that contribute to the reduction of cyberbullying.

Toward this end, we conducted three scenario-based experiments. The first experiment investigates whether displaying IP locations can mitigate cyberbullying within the context of Douyin, a well-known online social platform in China. The second experiment delves into the mediating role of impression management motivation and the moderating influence of group identity within the context of Bilibili, another Chinese social platform, to further understand how IP location display affects cyberbullying. Furthermore, previous research has found that internet anonymity can facilitate individuals' cyberbullying behavior by increasing moral disengagement¹⁶. Other studies have also suggested that internet anonymity will reduce users' online privacy concerns¹⁷, and in turn, increase their engagement in cyberbullying¹⁸. Therefore, we conducted the third experiment in the scenario of another major social media platform, Sina Weibo, to validate the findings from the first two experiments, while controlling for the potential explanatory mechanisms of moral disengagement and privacy concerns. Our three studies adopt a stepwise, cumulative design: each builds on its predecessor and draws on a different social media platform. We anticipate that this cross-platform triangulation will enhance the robustness and external validity of the findings.

This study contributes both theoretically and practically in the following ways. Firstly, we propose a novel approach to alleviate cyberbullying by disclosing users' IP locations on social platforms and verify its effectiveness. Unlike previous cyberbullying prevention measures, this study adopts an approach from the perspective of optimizing platform design, overcoming the limitations of high costs, limited coverage, and low flexibility and controllability inherent in past measures^{4–8}. Secondly, we enrich the theoretical understanding of how internet anonymity influences cyberbullying by leveraging impression management theory. Different from the moral disengagement and privacy concern mechanisms mentioned in prior research^{17–19}, we reveal that displaying IP locations reduces cyberbullying by enhancing users' motivation for impression management, which is a strategy to prevent damage to one's image. Thirdly, we also take into account the group characteristics of online social platforms by identifying group identity as a contingent factor influencing the relationship between IP location display and cyberbullying, thereby enriching our understanding of the dynamics between the two. Lastly, our findings offer practical guidance for internet policymakers, social platform managers, and other individuals who may be involved in cyberbullying.

Theoretical background

Cyberbullying and internet anonymity

Distinct from traditional bullying behaviors, cyberbullying is characterized by anonymity, accessibility, and publicity¹⁹. It occurs more covertly, spreads more rapidly, and lasts longer²⁰, thus leading to more profound influences^{21,22}. The negative effects of cyberbullying can be reflected at the individual and social levels. On the one hand, it harms individuals' physical and mental health^{2,20,23}, manifested as depressive symptoms^{24–26}, emotional exhaustion²⁷, reduced well-being²⁸, decreased life satisfaction²³, perceived stress^{29,30}, social anxiety³, and even suicidal ideation^{2,29,31}. On the other hand, the widespread dissemination of cyberbullying also poses a significant threat to social harmony and stability. In the online environment, individuals are more likely to imitate the bullying behaviors of others, thereby intensifying the spread of cyberbullying²¹. They are also susceptible to group pressure, either coerced or voluntarily participating in bullying behaviors, leading to the formation of collective bullying phenomena³². Additionally, cyberbullying is often accompanied by intense negative emotions such as anger, fear, and disgust. These emotions rapidly propagate in cyberspace, further encouraging more people to join in cyberbullying³³.

To curb cyberbullying, prior research has examined psychological mechanisms for reducing such behavior, which can be broadly grouped into three categories. One category is value and affective systems, including empathy^{34–37} and guilt and moral identity³⁸, which inhibit aggression by reducing the rationalization of harm.

The second category is self-regulatory systems, including self-control^{39,40} and mindfulness⁴¹, which reduce aggression by lowering impulsive and hostile responses. The third category is social mobilization systems, including norm clarification²¹, accountability cues^{42,43}, and bystander self-efficacy⁴⁴, which reduce cyberbullying through perceived oversight. This perceived monitoring may trigger a chilling effect and privacy strain. For example, in the Weibo context, research on the mandatory display of IP location finds that many users perceive costs to outweigh benefits in their privacy calculus, thereby reducing self-disclosure and adopting stronger privacy management strategies⁴⁵. Such privacy concerns further suppress users' harmful expression⁴⁶.

Industry and academia have exerted considerable effort in exploring effective ways to prevent cyberbullying. Firstly, the tendency for people to engage in cyberbullying can be suppressed through training and education. For instance, serious games and online courses related to anti-cyberbullying can help create empathy for victims, raise awareness of the consequences of cyberbullying, and teach strategies to combat cyberbullying^{4,47}. Additionally, brand advertisements can also prompt consumers to support anti-cyberbullying activities by reminding them that they have acted contrary to their moral values⁴⁸. Yet, training and education often require significant investment and can only reach a limited audience. Secondly, machine learning and other algorithmic models are used to detect malicious comments and risky users, thereby filtering out cyberbullying content^{6,7}. The strategy relies on continuous data collection and algorithm improvement to ensure accuracy. However, the speed of cyberbullying occurrence and dissemination often exceeds the speed of detection and reporting⁴⁹, rendering the algorithmic models sometimes ineffective. Thirdly, internet bystanders can also play a role in inhibiting cyberbullying. Research shows that when cyberbullying occurs, bystanders are willing to protect the victims^{5,8}. Therefore, encouraging bystanders to monitor and intervene in online behavior can help reduce cyberbullying. However, this approach does not address the root cause of the problem and has relatively low controllability.

In contrast, optimizing the functional design of user-oriented social platforms may be a more convenient, controllable, and cost-effective solution to alleviate the problem of cyberbullying. For instance, the Chinese government mandates social platforms to display the IP locations of all users. Since 2022, Chinese social media platforms have simultaneously displayed users' IP locations on the home page and in interaction areas, creating a semi real name environment with geographic identity cues. Existing research generally treats this policy as an institutional change in the information environment to examine how identity cues affect online behavior and the discursive ecology⁴⁵. These studies argue that IP location display elicits users' privacy concerns⁴⁵, promotes topic territoriality⁵⁰, fosters privacy cynicism and reduces perceptions of long-term privacy threats⁵¹, and shapes users' perceptions of the geography of digital space⁵². Building on this literature, the connections and differences between this study and prior work lie in two points. First, this study advances research on IP location display and user psychology and behavior to the governance of antisocial behavior by focusing on the suppressive effect of IP location display on cyberbullying. Second, although prior studies show that IP location display heightens users' privacy concerns and leads to more cautious privacy management strategies⁴⁵, they have not tested whether this process can curb harmful online behavior. This study treats privacy concern as one potential psychological mechanism and incorporates it together with moral disengagement and impression management motivation into a unified framework to identify the dominant psychological mechanism through which IP location display reduces cyberbullying.

This study contends that making IP locations public is a means of weakening online anonymity. The integration of IP geolocation with social platforms is constituting a new "spatial medium" that reshapes the spatial sense of online interaction and its normative boundaries, thereby enhancing the visibility of responsibility and affiliation cues⁵². Previous research has shown that anonymity affects users' online behavior through the mechanisms of disinhibition and deindividuation⁵³. First, the disinhibition effect refers to the reduction of self-monitoring and self-restraint due to perceived anonymity in the online environment, leading to more open or aggressive behavior⁵⁴. Second, deindividuation means the loss of self-awareness and reduced sense of personal responsibility in a group, making individuals more likely to engage in antisocial or non-normative behavior¹⁸.

Furthermore, the relationship between online anonymity and cyberbullying unfolds along two individual level psychological branches. The first is moral disengagement: online anonymity and the resulting reduced accountability and disinhibition weaken moral self-regulation, making individuals more likely to justify harmful expression with claims such as "the victim does not deserve sympathy" or "it is only a joke," thereby increasing the likelihood of perpetration. For example, platform features can activate moral disengagement through experiences of deindividuation, which in turn encourages bystanders to join cyberbullying¹⁹. Following this logic, this study argues that when identifiability and accountability cues are stronger, the psychological bases of diffusion of responsibility and deindividuation are weakened, making it more difficult for individuals to seek justification for aggressive behavior through moral disengagement, and therefore easier to suppress cyberbullying.

The second is privacy concern: when platforms reduce online anonymity and strengthen identifiable cues, concerns about traceability and exposure increase, producing self-censorship and avoidance tendencies. This chilling effect has been observed in multiple online environments^{46,55}. However, direct evidence remains limited that reductions in online anonymity decrease cyberbullying by heightening privacy concern. Moreover, long term monitoring can reduce sensitivity to privacy threats and foster greater ability to neutralize the impact of surveillance on oneself, which means that from a long-term perspective privacy concern may no longer be an effective psychological mechanism between online anonymity and cyberbullying⁵¹. At the same time, these two mechanisms reflect more passive psychological responses that arise for individuals in the environment. Moral disengagement emphasizes a failure of internal moral self-regulation, and privacy concern emphasizes self-protection in the face of risk. This differs from impression management motivation, which emphasizes active compliance driven by social evaluation goals and audience salience and represents an active adjustment of psychological state. It is therefore necessary to discuss impression management motivation as a complementary psychological mechanism and to further determine which of the three mechanisms plays the dominant role between online anonymity and cyberbullying.

On this basis, this study introduces impression management motivation as a third individual mechanism. A large body of research indicates that impression management is a pervasive and enduring motivation in interpersonal interaction. Individuals in most social situations continuously attend to how others perceive them and adjust self presentation accordingly^{13,56}. In social media environments, this enduring motivation is continually activated through imagined audiences and selective self presentation⁵⁷. Identity cues and accountable environments increase audience salience and the visibility of social evaluation, prompting individuals to actively regulate overt behavior to maintain face and reputation, thereby reducing the incidence of rule violations and aggressive expression.

Given the current lack of empirical evidence on how anonymity reducing IT designs influence users' psychology and, in turn, their participation in cyberbullying, we next draw on impression management theory to examine the underlying mechanisms.

Impression management theory and group identity

Impression management theory originated from the field of social psychology⁵⁶. The theory posits that people play different roles in various social situations and engage in behaviors corresponding to these contexts, to present the desired image to other audience members. Based on this, Leary and Kowalski divided the impression management process into two subprocesses: impression management motivation and impression construction¹³. Impression management motivation describes an individual's desire to engage in impression management behaviors, while impression construction refers to the strategies individuals adopt for impression management. There is a causal relationship between these two subprocesses, meaning that people need to activate their impression management motivation before engaging in impression management behaviors.

Gardner and Martinko argued that impression management theory drew inspiration from and absorbed the interaction of the three key elements of social learning theory, namely individual, behavior, and environment⁵⁸. Specifically, performers make judgments and definitions about situations based on their personality traits, cognitive experiences, and environmental characteristics, and then engage in corresponding impression management behaviors. Similarly, audiences also make judgments about the current situation based on their personality traits, cognition, and environmental factors, and react to the impressions created by the performers after observation. When an individual's actions are more observable by others, they become more concerned with how their actions influence others' opinions, thus having a stronger motivation to engage in impression management behaviors^{13,59,60}. Impression management behaviors can be categorized into two types: assertive impression management and defensive impression management⁶¹. Assertive impression management involves taking actions to enhance one's own impression in order to gain social recognition, whereas defensive impression management aims to maintain existing impressions and avoid impression damage⁵⁸.

In addition, scholars have discovered a link between the lack of anonymity and impression construction⁶². Specifically, when the anonymity of the environment is low, individuals will increase their impression management motivation, and may adopt relevant impression strategies to safeguard their image. Conversely, the high anonymity of the internet environment makes it difficult to observe people's true identities, leading to a diminished motivation for impression management and an increased likelihood of engaging in deviant behaviors⁶³. Therefore, social platforms disclosing IP locations to reduce anonymity may prevent users from participating in negative activities that damage their impression, such as cyberbullying, by increasing their motivation for impression management. Impression management theory provides new insights into studying the relationship between IP display and cyberbullying.

Specifically, this study treats IP location display as a publicness and accountability cue that reduces anonymity, whose effect on cyberbullying occurs indirectly through impression management motivation. The mechanism involves three sequential stages. First, IP display lowers actors' subjective judgments of their own anonymity, and reduced online anonymity elevates public self-awareness and perceived accountability⁶⁴. Second, this visibility and expectation of accountability strengthens individuals' impression management motivation, making them more inclined to invest in self-presentation and reputation maintenance, whether through assertive strategies to gain approval or defensive strategies to avoid image loss¹³. Third, because cyberbullying clearly violates social norms and platform rules and entails high risks of negative reputation and sanctions, it is more likely to be actively suppressed once impression management motivation is activated, thereby reducing its occurrence²¹.

Furthermore, the motivation for impression management is closely linked to an individual's identity. People are sorted into various groups based on shared physiological traits, attitudes, interests, and goals⁶⁵. An individual's self-concept encompasses both personal identity, which pertains to beliefs about one's own skills and unique attributes, and social identity, which relates to the groups and social categories the individual belongs to⁶⁶. Groups serve as a source of individuals' understanding of self-worth⁶⁷. When group-related beliefs significantly influence an individual's self-concept, this indicates a strong group identity⁶⁸, which in turn can influence attitudes and behaviors¹⁵. This dynamic is observed in both established and newly formed social groups⁶⁹. When individuals possess a strong group identity, they tend to see themselves as integral members of their respective groups⁷⁰. They develop more positive attitudes towards the group⁷¹, and are concerned with how their group is perceived by external environments and other groups^{72,73}. Consequently, they are likely to engage in behaviors conducive to the group to maintain a positive evaluation of the group⁷⁴. In other words, a high level of group identity triggers an individual's motivation to engage in impression management to maintain the group's image. In the context of this study, online social platform users are quickly categorized into different geographical groups based on their IP locations, which will also facilitate the formation of group identity.

Specifically, at the behavioral level of social media, geographic segmentation is not a weak cue. Using Twitter as an example, large scale data show that social ties and interactions display strong geographic homophily, with within city and within region connections markedly denser, and distance and boundaries clearly shaping online network structure. This indicates that geographic cues have stable explanatory power in online interaction⁷⁵.

At the same time, this study focuses on province level IP location display. In the Chinese context, provinces themselves carry meaningful cultural and institutional differences that may amplify or suppress aggressive expression. For example, differences between rice and wheat cultivation rooted in agricultural history reveal psychological and social behavioral differences across Chinese provinces⁷⁶.

In line with prior theoretical logic on group identity, such simple large groups can induce similar behavioral responses among members, and even when individuals are passively assigned to different groups, they can quickly develop a preference for their own group⁷⁷. Even if individuals move to a place solely for work and are not natives of that location, their social psychology may still be contextually influenced by the place of employment, even when this influence arises only from the connection to work. For example, an individual from Beijing who works in Shanghai may develop a sense of belonging to Shanghai because it is perceived as more youthful and fashionable than Beijing, or due to local culinary preferences. Therefore, group identification is not necessarily related to being a native of a place. Even when relocating only for work, individuals may develop a strong identification with the work locality, or they may develop aversion to it because of work experiences. These differences can be reflected in measurements of individual group identification.

Consequently, group identity serves as a significant contingent factor influencing the relationship between IP disclosure and users' impression management motivation and behaviors.

Research model and hypotheses

Drawing on impression management theory, this study developed a theoretical model encompassing six hypotheses, as shown in Fig. 1. Initially, displaying IP location is posited to have a deterrent effect on cyberbullying. Moreover, the motivation for impression management serves as a mediating factor in the relationship between IP location display and cyberbullying, while controlling for the alternative mechanisms of moral disengagement and privacy concerns. The visibility of IP location is expected to intensify individuals' motivation to manage their impressions, which in turn, is associated with a decrease in cyberbullying behavior. Lastly, group identity is proposed to act as a positive moderator between IP location display and impression management motivation. It is suggested that group identity enhances the influence of IP location display on cyberbullying behavior through the lens of impression management motivation. Next, we will elaborate on these hypotheses. The mediation effect of H4 and the moderated mediation effect of H6 are not displayed in the figure. H4: Impression management motivation mediates the relationship between IP location display and cyberbullying behavior. H6: Group identity positively moderates the influence of IP location display on cyberbullying behavior through impression management motivation. The dashed line represents the explanatory mechanism that we need to control.

Almost at the beginning of the birth of the network, it was accompanied by anonymous attributes. When registering an account, people can use false names, gender, and age, steal other people's pictures for online dating, or cover up the details that may reveal their real identity in the pictures⁷⁸. Because of this, some individuals' online behavior began to become reckless, especially those with rebellious psychology. In the real

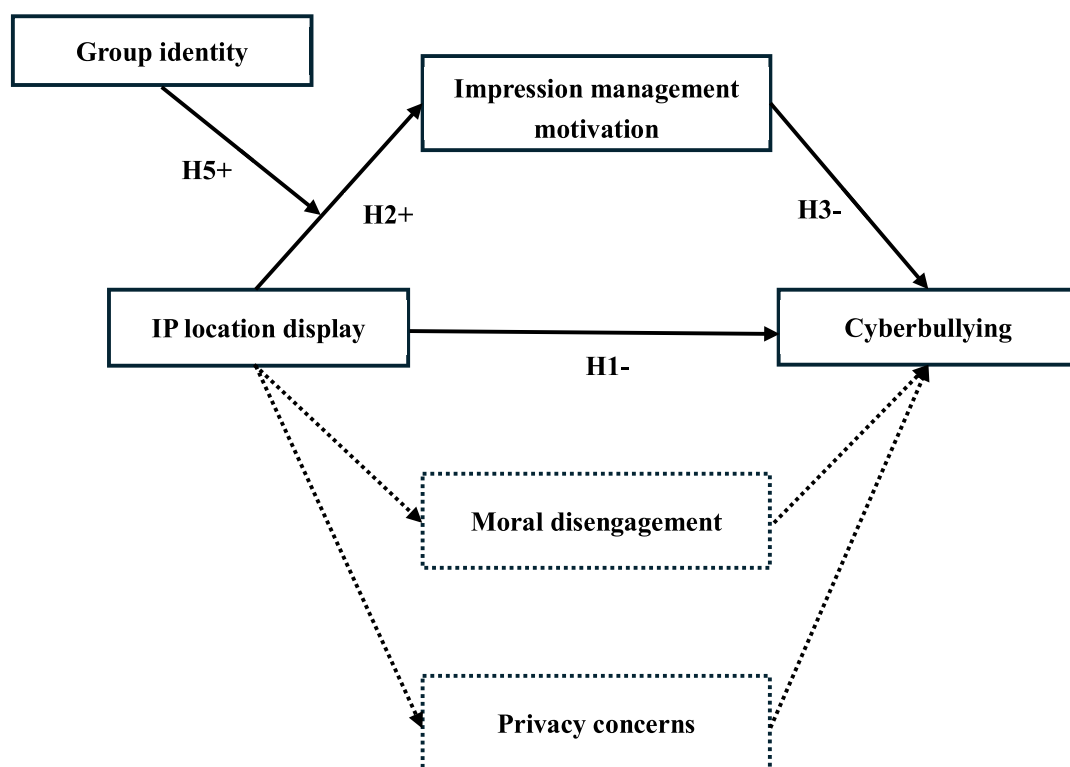


Fig. 1. Theoretical model.

world, they may restrain their behavior due to the constraints of social norms. But under the protection of internet anonymity, even if people who engage in cyberbullying will be criticized by the crowd, they know that this is only for a virtual image, since it is difficult to associate with their real identity. People only need to change a new account to escape from social monitoring, which means people don't need to pay for cyberbullying.

On the contrary, because the IP location display affects the internet anonymity by revealing clues about the real identity of the individual behind the account, when the IP location is displayed, the internet anonymity is reduced. No matter how many accounts the individual switches, the personal home page, and comments will display the real information of the geographical location of the individual in the provincial category, so the IP location information becomes a clue to reflect the real world and build a bridge to connect the virtual world. However, internet anonymity is related to impression construction⁷⁹. According to impression management theory, impression construction refers to the way in which individuals conduct impression management¹³, which involves individuals' choice of what type of information to present¹⁰, such as restraining their negative behavior to cut off the transmission of information that is not conducive to them, especially when individual identity is easier to be identified. The inhibition of cyberbullying is to reduce the loss of individual impressions through control, which can be regarded as a way of impression construction. Furthermore, among many impression management behaviors, defensive impression management is a kind of strategy to minimize the loss of impression and to maintain the existing image⁵⁸. When individuals perceive that their personal image is threatened, they usually adopt defensive strategies^{58,80}. When the IP location is displayed, the anonymity of the network is reduced, and the possibility of individuals' perception of the threat to their real image is increased. In order to maintain the existing image, they decide to adopt the defensive impression strategy, which means they will control their tendency to engage in cyberbullying.

Additionally, the image displayed by people is affected not only by self-expectation, but also by the expectation of others¹³. Online social networks are a platform for individuals to observe and learn. In the process of observation, individuals can find that users engaged in cyberbullying will be strongly criticized by other individuals. This is because cyberbullying, as an act of undermining social order, reduces the perceived similarity of the public to the perpetrators, and the perpetrators are considered to be unsocial and unwelcome, resulting in a loss of impression on the perpetrators⁵⁸. Such observation constructs the causal logic between individual impression management strategies and others' reactions. This logic may be weak in the context of strong internet anonymity, because people know that changing a new account is to give themselves opportunities to rebirth. However, the display of the IP location reveals clues about the real identity behind the account. People can infer the possible loss of impression in the future through this logic, so as to prevent the real damage to the image by predictably suppressing cyberbullying, which shows that people are more cautious in considering and following this causal logic compared to anonymous situations. Therefore, we propose the following hypothesis:

H1: IP location display has a negative effect on cyberbullying behavior.

According to the two-component model of impression management proposed by Leary and Kowalski, impression management motivation refers to the desire of individuals to create a certain impression¹³. Before people take specific impression management behaviors, they will first generate impression management motivation. As for the activating factors of impression management motivation, Goffman believes that environmental factors affect people's judgment of the situation⁵⁶, such as the physical structure, objects, and signs within the organization⁵⁸. Similarly, the display of IP location represents a sign of cyberspace, which affects people's judgment as environmental clues.

At the same time, people's judgment of the environment can be expressed as evaluative potential. Evaluative potential describes the degree to which people can feel the evaluation of others in the situation¹⁰. When people realize that the situation is vulnerable to the evaluation of others, they are more interested in their outward image, which also means higher motivation for impression management¹³. Furthermore, compared with the private context, the public context can make others observe the individual's behavior more easily^{81,82}, so that it can make individuals feel the evaluation of others, and the more motivated they are to carry out impression management^{59,60,83,84}.

Meanwhile, the source of impression management motivation is similar to that of other behavioral motivations, that is, maximize expected reward and minimize expected punishment, which means that the punishment caused by the failure of impression management will enhance people's impression management motivation⁸⁵. When the IP location is displayed, the anonymity of the network is reduced. The reduction of anonymity makes people feel that their real identity is exposed to a more public environment and feel the gaze of social evaluation and supervision. In this case, people realize that engaging in cyberbullying will lead to the failure of impression management, thus arousing people's awareness of impression management to minimize their possible losses. Thus, we hypothesize:

H2: IP location display has a positive effect on impression management motivation.

The enhancement of impression management motivation has also been confirmed to be related to more positive behaviors, such as corporate social responsibility activity participation⁸⁶, creative process engagement⁸⁷, leadership emergence⁸⁸, organizational citizenship behavior⁸⁹, etc. It is because individuals construct their impression of a destination. Rosenberg and Egbert regard impression management as a goal-oriented behavior⁹⁰, and introduce Dillard et al.'s related concepts of interpersonal goals, including primary goals and secondary goals⁹¹. In the process of impression management, the primary goal is to create an ideal impression. The secondary goals are divided into self-oriented goals and interaction-oriented goals. Among them, self-oriented goals are related to individuals' desire to grow or maintain resources, and interaction-oriented goals are related to maintaining good social communication⁹⁰. Since cyberbullying is an insulting attack against others, it violates the moral norms advocated by society and will be condemned by other individuals or groups, which is not conducive to the smooth progress of social interaction. Furthermore, it affects the reputation of individuals, which can be regarded as their relationship resources. Therefore, for individuals whose impression management

motivation has been activated, whether based on interaction-oriented goals or self-oriented goals, individuals will try not to cyberbullying to avoid creating negative impressions. Based on this evidence, the following hypothesis is put forward:

H3: Impression management motivation has a negative effect on cyberbullying behavior.

To sum up, people's judgment of the current situation is the main reason that affects the follow-up impression management behavior. As the environmental clues of cyberspace, when the IP location is displayed, the anonymity of the network is reduced, and the logic between individuals' impression management strategy and others' response is strengthened. The individuals are prompted that the context is more public, which makes individuals perceive the vision of social supervision and evaluation. In this situation, out of fear of negative evaluation of real identity in the real world, individuals' motivation for impression management is enhanced. After that, when the motivation of individual impression management is activated, for the protection of individual reputation and the successful move of social communication, individuals tend to take defensive impression construction, that is, to curb cyberbullying. Impression management motivation plays an important intermediary role in the process of IP location display to the role of cyberbullying. Hence, we propose that:

H4: Impression management motivation mediates the relationship between IP location display and cyberbullying behavior.

Group identity represents the degree to which an individual self conceptualizes the group they are in⁶⁸. Individuals with high group identity, prefer the group they belong to⁷¹, have a stronger sense of group belonging, and pay more attention to external evaluation^{72,73} and benefit of the group⁹². Because individuals with high group identity have deeply integrated group labels into their self-concept, if the group is negatively evaluated by outsiders, they will not only consider it a shame to the group, but also consider it a shame to themselves, so that their self-esteem will be challenged. Therefore, high group identity individuals make efforts to maintain the positive external image of the group, even if these efforts are not specifically prompted by the group^{93,94}.

In China, the display of IP location does not reveal a string of numeric codes, but the name of the province to which the individual belongs. For example, users who are currently in Sichuan Province will display their IP location as Sichuan Province when commenting. Therefore, IP location display not only reduces internet anonymity by disclosing clues of personal real identity, but also divides individuals into groups according to 34 provinces in China, which is a way to divide groups based on demographics in the concept of group. Additionally, in the context of Chinese traditional culture, individuals often have deep emotional ties with their places of belonging, and countless literary works have expressed homesickness before, which symbolizes that geographical groups might have a certain impact on individuals.

According to the impression management theory, individual differences will make some kinds of people more sensitive to situational factors⁹⁵. In this study, individuals with high group identity are more sensitive to IP location display which is a kind of situational factor. When individuals are classified into different geographical groups by IP location information, individuals with high group identity show a strong emotional connection to the place they belong to. People know that once they are labeled as a group member, their words and deeds will represent the will of the group and impact the benefits and evaluation of it. Since they don't want the image of their place of belonging to be belittled by outsiders, even if the group does not officially require them to obey the social norms, individuals with high group identity still have stronger motivation to regulate their external image. As such, we hypothesize:

H5: Group identity positively moderates the influence of IP location display on impression management motivation.

People will judge the situation first, and influence the performance of their group to the outsiders by adjusting their personal behavior as group members⁵⁶. Specifically, when the individual and the group image have a closer connection, the individual is more likely to take the corresponding impression management behavior for the group image. IP location display promotes the binding of individual images with the image of their group, which is more motivating for individuals with high group identity to engage in personal impression management to maintain group image, and then adopt defensive impression construction to suppress cyberbullying behavior. Therefore, we propose the following hypothesis:

H6: Group identity positively moderates the influence of IP location display on cyberbullying behavior through impression management motivation.

Methodology

We conducted three scenario-based online experiments to explore how IP location display influences cyberbullying behavior. The online approach can increase participants' perception of anonymity and mitigate the social desirability bias that often arises when measuring negative behaviors⁹⁶. Specifically, Experiment 1 tested the hypothesized effect of IP location display on users' cyberbullying behavior (H1). Experiment 2 examined users' impression management motivation as an underlying mechanism (H2-H4) and group identity as a potential boundary condition (H5-H6). Finally, Experiment 3 replicated the results of Experiments 1 and 2 and ruled out two key alternative explanations.

Experiment 1

Sample and procedure

The main purpose of Experiment 1 was to examine the main effect of IP location display on cyberbullying behavior. We recruited participants on Douyin, one of the most popular social media platforms in China, with more than 750 million daily active users (<https://www.businessofapps.com/data/tik-tok-statistics/>, accessed 10 October 2024).

We first introduced the experimental objectives and response requirements to the participants, and assured them of the anonymity and confidentiality of the data that would be collected⁹⁶. After the participants confirmed

their continued participation, we asked them to imagine a scenario where they are using Douyin, engaging in activities such as liking, commenting, sharing, and sending private messages. We then randomly assigned the participants to two groups in a between-subjects design: IP Display group and IP Non-Display group. In the IP Display group, participants were informed that “the platform requires the display of user IP location, so that other users can see your IP location on your homepage or when you leave comments.” Conversely, in the IP Non-Display group, participants were told that “other users cannot see your IP location on your homepage or when you leave comments.” To enhance their understanding, we also presented images showing personal homepages with and without the IP display (see Fig. 2).

After presenting the scenario, we used a manipulation check question to ensure the successful manipulation of IP display. Participants were asked to recall whether others could see their IP location while they were using Douyin in the imagined scenario. Those who failed the manipulation check were terminated from the experiment. Then, participants responded to the likelihood of their engaging in cyberbullying behavior, with seven items adapted from Chan et al. (2019) and Lowry et al. (2019) (“I may post hurtful, rude, inappropriate, or mean content that targets someone”; 1 = strongly disagree, 7 = strongly agree, $\alpha = 0.941$)^{97,98}. In addition, they completed a series of demographic questions, including age, gender, education level, income, and daily use duration of Douyin. Interspersed among these questions, we also included five attention check questions (e.g., “The sun rises in the east and sets in the west”). Those who answered incorrectly were excluded from subsequent analyses. Finally, we thanked the participants and distributed the compensation. Out of the 534 participants, 378 completed the experiment and passed the attention checks, resulting in a response rate of 70.78%. In the end, there were 192 samples in the IP Display group and 186 in the IP Non-Display group.

Results

We employed both analysis of variance (ANOVA) and multiple regression to validate our hypotheses. One-way ANOVA indicated a significant difference in cyberbullying behavior between the two groups ($\eta^2 = 0.029$, $F = 11.190$, $p < 0.001$). Specifically, the mean value of cyberbullying in the IP Display group ($M_{\text{display}} = 1.469$, $SD_{\text{display}} = 0.684$) was lower than that in the IP Non-Display group ($M_{\text{non-display}} = 1.825$, $SD_{\text{non-display}} = 1.283$), indicating that displaying IP location could significantly reduce cyberbullying behavior. Thus, H1 was supported. Figure 3 depicts the visualized results.

In addition, the regression results in Table 1 showed that IP location display had a negative influence on cyberbullying ($\beta = -0.335^{**}$, $p < 0.01$), also supporting H1.

Experiment 2

Sample and procedure

Experiment 2 aimed to further investigate the mediating role of impression management motivation and the moderation role of group identity in the relationship between IP location display and cyberbullying behavior. We chose another well-known social media platform in China, Bilibili with daily active users of 102.3 million, as the experimental setting to recruit participants (<https://ir.bilibili.com/media/qcmvd2g/bilibili-inc-announce-second-quarter-2024-financial-results.pdf>, accessed 30 June 2024).

The process of Experiment 2 was essentially the same as that of Experiment 1, but the scenario in the experimental materials was set on Bilibili. Furthermore, besides cyberbullying behavior ($\alpha = 0.906$) and demographic information, the participants also responded to questions regarding impression management motivation and group identity. The measurement of impression management motivation (“I desire to create a good impression when communicating with others on Bilibili”; 1 = strongly disagree, 7 = strongly agree, $\alpha = 0.703$) was adapted from the supervisor-attributed impression management motives scale developed

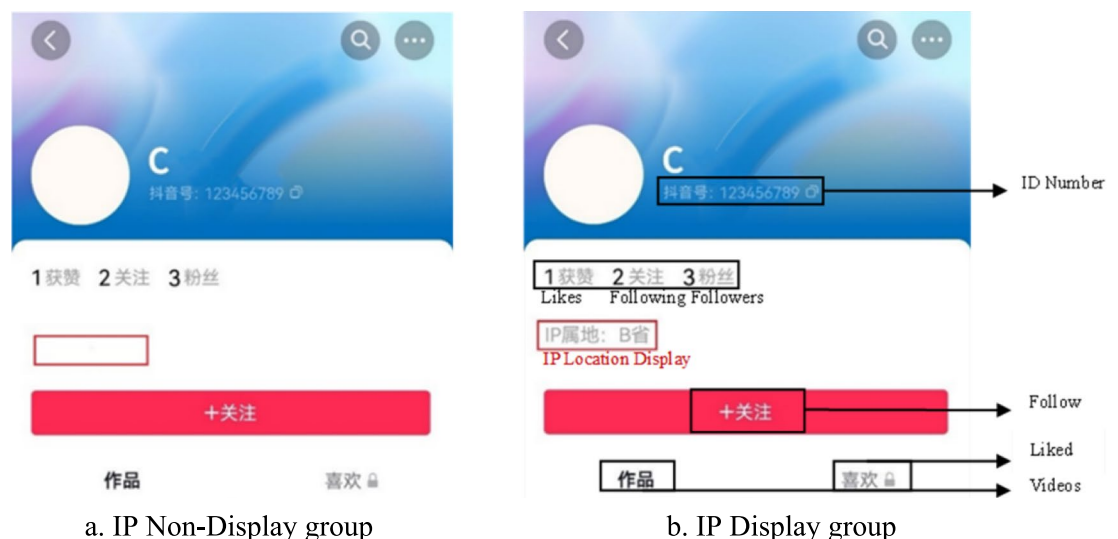


Fig. 2. Personal homepages of Douyin.

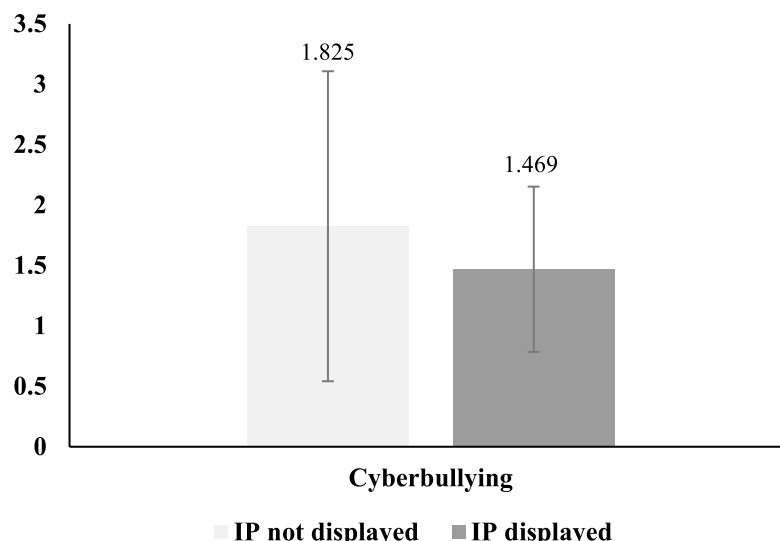


Fig. 3. Cyberbullying by condition in Study 1.

Variable	Cyberbullying
IP location display	−0.335** (0.002)
Gender	0.022 (0.840)
Age	0.092 (0.184)
Education	0.132 (0.141)
Income	−0.094 (0.175)
Duration	0.077 (0.201)
R ²	0.044
F	2.818*

Table 1. Multiple regression results for Study 1. Unstandardized coefficients (*p*-values) are reported. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; two-tailed test.

by Lam et al. (2007)⁹⁹, removing two items that were not applicable to the current scenario. The six items of group identity (“When someone criticizes the city I am in, it feels like a personal insult”; 1 = strongly disagree, 7 = strongly agree, $\alpha = 0.803$) were adapted from Mael & Ashforth’s (1992) organizational identification scale¹⁰⁰. Ultimately, we retained 624 valid samples from 867 participants, with a response rate of 71.972%. Among them, 340 were randomly assigned to the group without IP display, and 284 were assigned to the group with IP display.

Results

Direct effect The results of one-way ANOVA indicated that there were significant differences between the two groups in terms of both cyberbullying behavior ($\eta^2 = 0.074$, $F = 56.144$, $p < 0.001$) and impression management motivation ($\eta^2 = 0.048$, $F = 32.500$, $p < 0.001$) (Fig. 4). The IP Display group ($M_{\text{display}} = 1.411$, $SD_{\text{display}} = 0.469$) exhibited lower levels of cyberbullying behavior compared to the IP Non-Display group ($M_{\text{non-display}} = 1.905$, $SD_{\text{non-display}} = 1.100$), supporting H1. The impression management motivation of the IP Display group ($M_{\text{display}} = 6.076$, $SD_{\text{display}} = 0.654$) was higher than that of the IP Non-Display group ($M_{\text{non-display}} = 5.748$, $SD_{\text{non-display}} = 0.785$), validating H2.

The regression results in Table 2 suggested that IP display had a negative effect on cyberbullying (Model a, $\beta = -0.498^{***}$, $p < 0.001$), also providing evidence for H1. Furthermore, IP display positively influenced impression management motivation (Model b, $\beta = 0.325^{***}$, $p < 0.001$), and impression management motivation negatively influenced cyberbullying (Model b, $\beta = -0.259^{***}$, $p < 0.001$), supporting H2 and H3, respectively.

Moderation effect Next, we used Process Macro Model 1 in SPSS 26.0 to conduct 5000 bootstrap resampling to test the moderation effect. The results of Table 2 indicated that the interaction between IP location display and group identity had a significant positive effect on impression management motivation (Model c, $\beta = 0.062^*$, $p < 0.05$), which meant that the higher the level of group identity among individuals, the stronger the positive impact of IP display on impression management motivation. Additionally, the interaction effect plot in Fig. 5 revealed that the impact of IP display on impression management motivation was significantly positive under high group identity, whereas insignificant under low group identity. Therefore, H5 was supported.

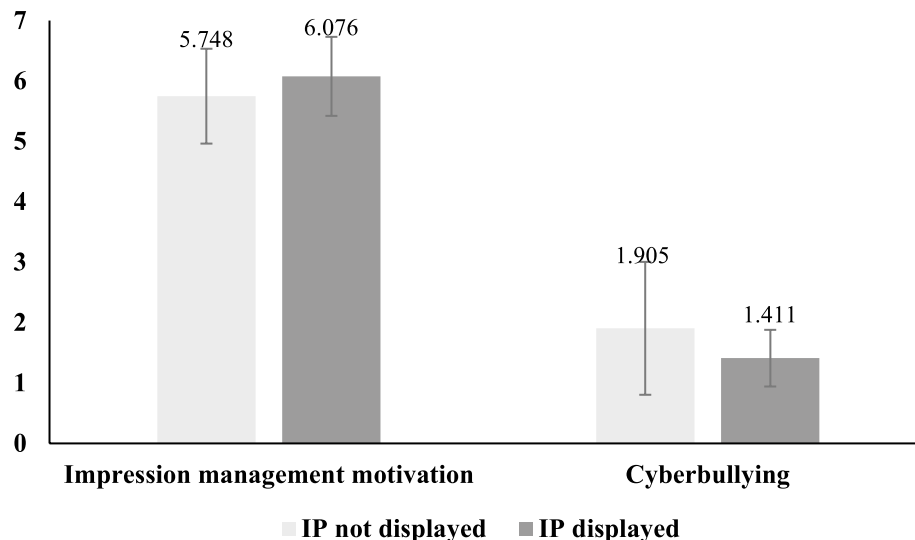


Fig. 4. Impression management motivation and Cyberbullying by condition in Study 2.

Variable	Model a. Direct effect	Model b. Mediation effect		Model c. Moderation effect	
	Cyberbullying	Impression management motivation	Cyberbullying	Impression management motivation	Cyberbullying
IP location display	−0.498*** (0.000)	0.325*** (0.000)	−0.413*** (0.000)	0.074** (0.005)	−0.206*** (0.000)
Impression management motivation			−0.259*** (0.000)		−0.259*** (0.000)
Group identity				0.385*** (0.000)	
IP location display * Group identity				0.062* (0.025)	
Gender	−0.072 (0.317)	−0.020 (0.737)	−0.077 (0.272)	0.002 (0.965)	−0.077 (0.272)
Age	0.141** (0.008)	−0.062 (0.159)	0.125* (0.015)	−0.029 (0.456)	0.125* (0.016)
Education	−0.037 (0.523)	0.007 (0.890)	−0.035 (0.533)	−0.001 (0.987)	−0.035 (0.533)
Income	0.057 (0.094)	0.050 (0.078)	0.070* (0.036)	0.030 (0.224)	0.070* (0.036)
Duration	−0.058 (0.115)	0.119*** (0.000)	−0.027 (0.456)	0.058* (0.031)	−0.027 (0.456)
R ²	0.102	0.082	0.144	0.309	0.144
F	10.048***	7.832***	12.967***	30.508***	12.967***

Table 2. Multiple regression results for Study 2. Unstandardized coefficients (*p*-values) are reported. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; two-tailed test.

Mediation effect We adopted Process Macro Model 4 to conduct 5000 bootstrap resampling to test the mediation effect. The results are shown in Table 3. The mediating effect of impression management motivation is -0.084 , with a 95% confidence interval of $[-0.140, -0.042]$, which does not include zero, indicating that impression management motivation significantly mediated the negative effect of IP display on cyberbullying behavior. Thus, H4 was supported.

Moderated mediation effect We performed a moderated mediation test using Process Macro Model 7 with 5000 bootstrap resamples. Table 4 showed that the index of moderated mediation was significant ($\beta = -0.016$, bias-corrected 95% confidence interval: $[-0.034, -0.001]$), indicating that group identity moderated the indirect effect of IP location display on cyberbullying behavior through impression management motivation. Specifically, the effect of IP location display on cyberbullying is significantly negative when group identity is high ($\beta = -0.035$, bias-corrected 95% confidence interval: $[-0.061, -0.016]$), and becomes insignificant when group identity is low ($\beta = -0.003$, bias-corrected 95% confidence interval: $[-0.028, 0.020]$). Hence, H6 was supported.

Experiment 3

Sample and procedure

The primary goals of Study 3 were to examine the findings of the previous two studies and to rule out alternative explanatory mechanisms such as moral disengagement and privacy concerns. The experimental setting was changed to Sina Weibo, a leading social media platform in China, which is a Twitter-like microblogging site

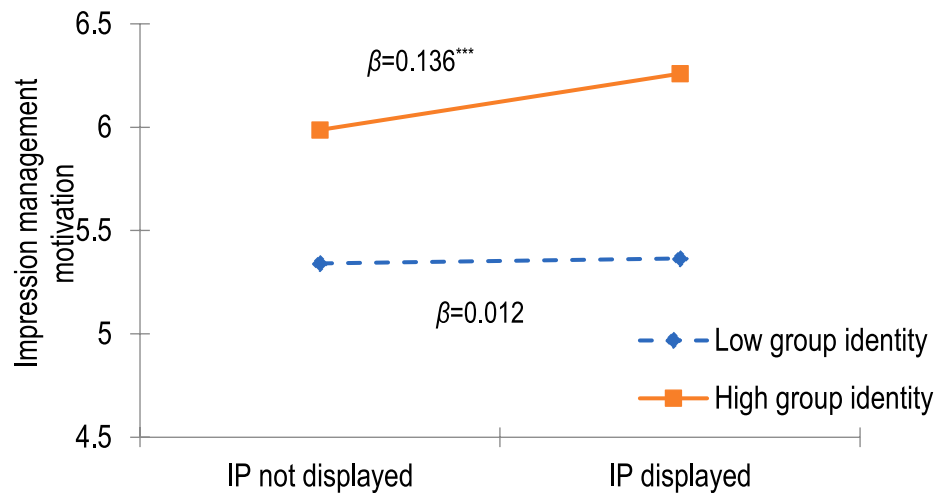


Fig. 5. Interaction plot for Study 2. Unstandardized coefficients (*p*-values) are reported. ***: *p* < 0.001; **: *p* < 0.01; *: *p* < 0.05; two-tailed test. — Significant----- Non-significant.

	Effect (SE)	Bias – corrected 95% confidence interval	
		Lower bound	Upper bound
Total effect	– 0.498 (0.069)	– 0.634	– 0.361
Direct effect	– 0.413 (0.070)	– 0.550	– 0.277
Mediation effect	– 0.084 (0.025)	– 0.140	– 0.042

Table 3. Mediation test results for Study 2.

Independent variable	Dependent variable	Conditional indirect effects through impression management motivation (SE)		Bias – corrected 95% confidence interval	
				Lower bound	Upper bound
IP location display	Cyberbullying	Low group identity	– 0.003 (0.012)	– 0.028	0.020
		High group identity	– 0.035 (0.012)	– 0.061	– 0.016
		Index of moderated mediation	– 0.016 (0.009)	– 0.034	– 0.001

Table 4. Moderated mediation test results for Study 2.

with about 256 million daily active users (<https://www.statista.com/statistics/1058070/china-sina-weibo-dau/>, accessed 2 September 2024).

Experiment 3 followed a process similar to Experiments 1 and 2, but the experimental scenario was set on Sina Weibo. In addition to questions about cyberbullying behavior ($\alpha=0.929$), impression management motivation ($\alpha=0.857$), group identity ($\alpha=0.849$), and demographic information, participants also answered questions related to moral disengagement and privacy concerns. We adopted the eight items for moral disengagement (“It is okay to spread rumors to defend those you care about”; 1 = strongly disagree, 7 = strongly agree, $\alpha=0.935$) from Moore et al. (2012)¹⁰¹. The five items measuring privacy concerns (“Writing negative review is risky, because other users of Sina Weibo may message me or contact me online”; 1 = strongly disagree, 7 = strongly agree, $\alpha=0.946$) were contextualized from the online privacy concerns scale by Zhu & Grover (2022)¹⁰². In the end, a total of 386 participants were recruited, and 268 valid samples were retained, yielding a response rate of 69.430%. Out of these, there were 118 samples in the IP Display group and 150 in the IP Non-Display group.

Results

Direct effect The results of one-way ANOVA revealed significant differences in both cyberbullying behavior ($\eta^2=0.079$, $F=28.191$, $p<0.001$) and impression management motivation ($\eta^2=0.105$, $F=37.878$, $p<0.001$) between the two groups. Figure 6 shows the visualized results. To elaborate, participants in the IP display condition ($M_{\text{display}}=1.697$, $SD_{\text{display}}=0.341$) showed a lower tendency for cyberbullying behavior than those in the IP non-display condition ($M_{\text{non-display}}=2.285$, $SD_{\text{non-display}}=1.299$). The impression management motivation of the IP Display group ($M_{\text{display}}=6.136$, $SD_{\text{display}}=0.501$) was higher than that of the IP non-display group ($M_{\text{non-display}}=5.431$, $SD_{\text{non-display}}=1.266$). These results indicated that both H1 and H2 were supported.

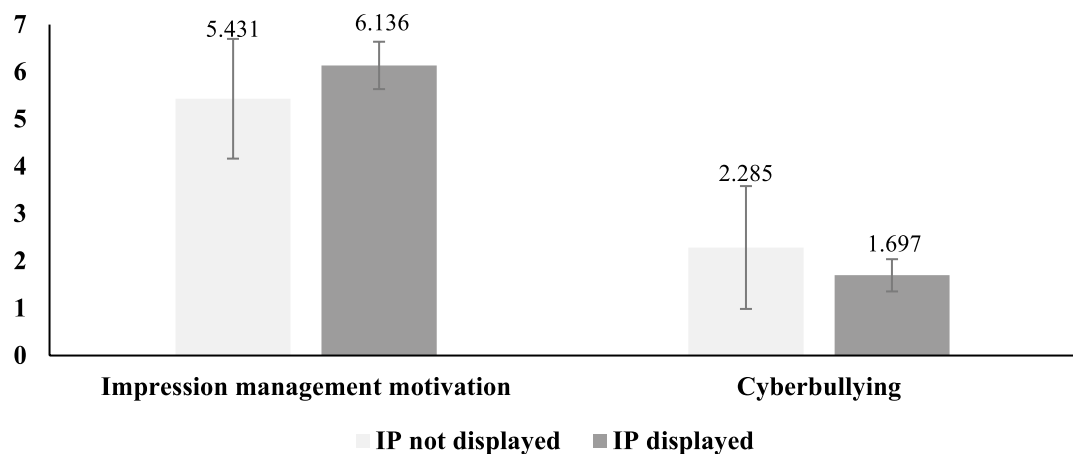


Fig. 6. Impression management motivation and Cyberbullying by condition in Study 3.

Variable	Model a. Direct effect	Model b. Mediation effect		Model c. Moderation effect	
	Cyberbullying	Impression management motivation	Cyberbullying	Impression management motivation	Cyberbullying
IP location display	−0.578*** (0.000)	0.695*** (0.000)	0.030 (0.706)	0.191*** (0.001)	0.015 (0.706)
Impression management motivation			−0.384*** (0.000)		−0.384*** (0.000)
Moral disengagement			0.456*** (0.000)		0.456*** (0.000)
Privacy concerns			−0.032 (0.222)		−0.032 (0.222)
Group identity				0.603*** (0.000)	
IP location display * Group identity				0.174* (0.045)	
Gender	−0.230 (0.064)	0.104 (0.402)	−0.094 (0.193)	0.155 (0.158)	−0.094 (0.193)
Age	−0.231* (0.019)	0.217 (0.029)	−0.015 (0.793)	0.183 (0.038)	−0.015 (0.793)
Education	−0.011 (0.927)	0.076 (0.548)	0.072 (0.322)	0.061 (0.584)	0.072 (0.322)
Income	−0.044 (0.491)	0.081 (0.207)	0.018 (0.632)	0.027 (0.642)	0.018 (0.632)
Duration	−0.031 (0.690)	−0.112 (0.159)	−0.075 (0.105)	−0.063 (0.367)	−0.075 (0.105)
R ²	0.123	0.148	0.711	0.345	0.711
F	5.198***	6.470***	63.064***	15.112***	63.064***

Table 5. Multiple regression results for Study 3. Unstandardized coefficients (*p*-values) are reported. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; two-tailed test.

The results of multiple regression analysis in Table 5 showed that IP location display negatively influenced cyberbullying (Model a, $\beta = -0.578^{***}$, $p < 0.001$), further validating H1. In addition, IP location display positively impacted impression management motivation (Model b, $\beta = 0.695^{***}$, $p < 0.001$), and impression management motivation exhibited a negative influence on cyberbullying (Model b, $\beta = -0.384^{***}$, $p < 0.001$), providing additional support for H2 and H3, respectively.

Moderation effect Next, we used Process Macro Model 1 in SPSS 26.0 to conduct 5000 bootstrap resampling to test the moderation effect. The results of Table 5 showed that group identity positively moderated the impact of IP location display on impression management motivation (Model c, $\beta = 0.174^*$, $p < 0.05$). Besides, Fig. 7 suggested that the impact of IP display on impression management motivation was significantly positive under high group identity, while insignificant under low group identity. These evidences collectively supported H5.

Mediation effect In order to rule out other potential mediating mechanisms, including moral disengagement and privacy concerns, we employed Process Macro Model 4 for the mediation effect analysis of multiple parallel mediating variables. Specifically, IP location display reduces cyberbullying behavior by enhancing impression management motivation ($\beta = -0.267$, bias-corrected 95% CI $[-0.437, -0.128]$) and weakening moral disengagement ($\beta = -0.294$, bias-corrected 95% CI $[-0.464, -0.154]$). But at the same time, privacy concerns do not play a mediating role in the relationship between IP location display and cyberbullying behavior ($\beta = -0.048$, bias-corrected 95% CI $[-0.118, 0.023]$). This indicates that after controlling for the mediating role of moral disengagement, impression management motivation still significantly mediated the negative impact of IP display on cyberbullying behavior, validating H4. The detailed results are presented in Table 6.

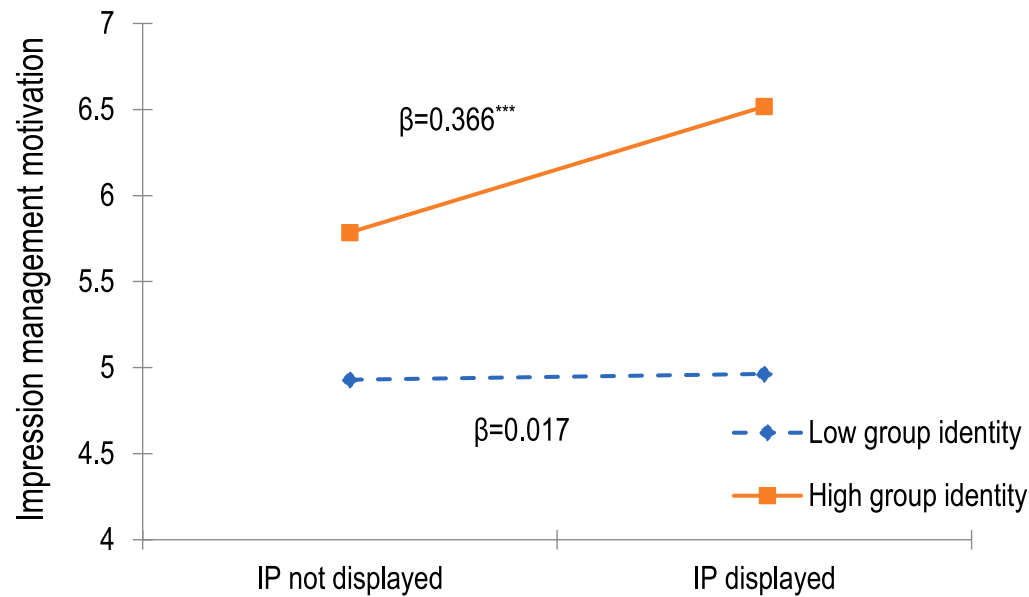


Fig. 7. Interaction plot for Study 3. Unstandardized coefficients (*p*-values) are reported. ***: *p* < 0.001; **: *p* < 0.01; *: *p* < 0.05; two-tailed test. — Significant----- Non-significant.

	Effect (SE)	Bias-corrected 95% confidence interval	
		Lower bound	Upper bound
Total effect	−0.578 (0.122)	−0.817	−0.338
Direct effect	0.030 (0.080)	−0.128	0.188
Mediation effect			
Impression management motivation	−0.267 (0.079)	−0.437	−0.128
Moral disengagement	−0.294 (0.080)	−0.464	−0.154
Privacy concerns	−0.048 (0.036)	−0.118	0.023

Table 6. Mediation test results for Study 3.

Independent variable	Dependent variable	Conditional indirect effects through impression management motivation (SE)		Bias-corrected 95% confidence interval	
				Lower bound	Upper bound
IP location display	Cyberbullying	Low group identity	−0.007 (0.035)	−0.077	0.063
		High group identity	−0.140 (0.050)	−0.253	−0.060
		Index of moderated mediation	−0.067 (0.033)	−0.140	−0.015

Table 7. Moderated mediation test results for Study 3.

Moderated mediation effect We further tested the moderated mediation effect by using Process Macro Model 7. The results in Table 7 showed that group identity moderated the negative effect of IP location display on cyberbullying behavior through impression management motivation (Index of moderated mediation = −0.067, bias-corrected 95% confidence interval: [−0.140, −0.015]). Specifically, the effect of IP location display on cyberbullying is significantly negative when group identity is high ($\beta = -0.140$, bias-corrected 95% confidence interval: [−0.253, −0.060]), and becomes insignificant when group identity is low ($\beta = -0.007$, bias-corrected 95% confidence interval: [−0.077, 0.063]). Therefore, H6 was supported.

Discussion

From the perspective of impression management theory, this study aims to discuss the impact of IP location display on cyberbullying, and the mediating role of impression management motivation, as well as the moderating role of group identity. Accordingly, this study conducted three experiments in total. The experiments were based on three well-known social network platforms in China, including Douyin, BiliBili, and Sina Weibo. The participants of these experiments included 378 Douyin users, 624 BiliBili users, and 268 Sina Weibo users.

The results show that: (1) Experiment 1 tests the main effect, indicating that IP location display has a negative effect on cyberbullying (H1); (2) Experiment 2 confirms the results of Experiment 1 (H1), and shows that impression management motivation mediates the impact of IP location display on cyberbullying. In particular, IP location display first improves individuals' impression management motivation, and the increased impression management motivation then suppresses their cyberbullying behavior (H2-H4). At the same time, Experiment 2 proves that group identity plays a moderating role in this process, which means that the stronger an individual's group identity is, the greater the positive impact of IP location display on impression management motivation, and consequently, the larger the inhibitory effect on cyberbullying (H5-H6). (3) Experiment 3 further verifies the conclusions of both Experiment 1 and Experiment 2 (H1-H6), and introduces two alternative explanatory mechanisms of moral disengagement and privacy concerns. The results show that IP location display could significantly lead to an increase in impression management motivation and privacy concerns, as well as a decrease in moral disengagement. However, the increase in privacy concerns did not further lead to a significant reduction in cyberbullying. In contrast, the two parallel mediating paths of impression management motivation and moral disengagement are established. After controlling for the mediating role of moral disengagement, the mediating effect of impression management motivation between the relationship of IP location display and cyberbullying remains significant, demonstrating the robustness of the results.

Implications for theory

Firstly, this study enriches the theoretical understanding of methods for controlling cyberbullying. Although people are fully aware of the serious consequences of cyberbullying and have developed measures such as psychological education, social support, algorithmic monitoring, and bystander intervention to curb it, these measures have their limitations. The display of IP location is a way to reduce the degree of anonymity by optimizing the functional design of user-oriented social platforms. Specifically, whether on personal homepages or when participating in comments, users can see the sign of IP location information, indicating the connection between their virtual and real identities. This study examines the effectiveness of this measure in our context. As the display of IP location is easier to operate and can cover a wider range of people, it forms an appropriate complement to other known methods of controlling cyberbullying.

Secondly, this study extends the discussion of IP location display and user psychology and behavior to the governance of antisocial behavior, focusing on its suppressive effect on cyberbullying. Since Chinese social media platforms began to display users' IP locations, four studies have examined its impact on user behavior. They report that IP display triggers privacy concerns⁴⁵, fosters topic territoriality⁵⁰, elicits privacy cynicism and a long term decrease in perceived privacy threat⁵¹, and shapes users' perceptions of the digital geography of online spaces⁵². These studies directly or indirectly suggest that IP display reduces online anonymity. Building on this foundation, the present study further demonstrates that the reduction in anonymity brought by IP display suppresses cyberbullying by increasing users' impression management motivation, and it extends the behavioral outcomes to antisocial behavior online. At the same time, although prior research shows that IP display heightens privacy concerns and encourages more cautious privacy management strategies⁴⁵, it has not tested whether this process can curb problematic online behavior. Accordingly, this study treats privacy concern as one potential psychological mechanism and incorporates it with moral disengagement and impression management motivation within a unified framework, thereby identifying impression management motivation as the dominant pathway through which IP display reduces cyberbullying.

Thirdly, this study adds a new theoretical explanation to the process from internet anonymity to cyberbullying. Previous studies on the relationship between internet anonymity and cyberbullying were mainly based on the theoretical perspectives of moral disengagement¹⁶, disinhibition^{53,103,104}, deindividuation^{18,53}, and social learning⁵³. These studies believed that more significant internet anonymity aggravates the distinction between personal real and virtual images, which reduces the inhibitory effect of social norms on individual behavior. Individuals realize that they may not need to be responsible for their mistakes, and the ambiguity of real identity gives them the opportunity to approach perpetrators, so that they can more easily imitate and learn bad behaviors, which ultimately leads to more cyberbullying. Although these studies have discussed the psychological process from internet anonymity to cyberbullying, they are concerned with the enhancement effect of this process, which can differ from the process of reducing internet anonymity to inhibit cyberbullying. While Cho & Kwon (2015) discussed the process of reducing internet anonymity to control cyberbullying⁹, it did not involve the change of individual psychological states. Therefore, this study introduces impression management theory as a new theoretical perspective to interpret the mechanism between internet anonymity and cyberbullying. The results show that impression management motivation and moral disengagement form two parallel psychological pathways under reduced anonymity, and the mediating effect of impression management motivation remains significant after controlling for moral disengagement, whereas privacy concern does not constitute an effective subsequent inhibitory pathway. According to Leary & Kowalski (1990)'s two-component model of impression management, we conceptualize IP location display as an environmental factor in the process of impression management, and not engaging in cyberbullying as an impression construction. Impression management motivation represents the psychological changes of individuals in this process¹³. The display of IP location means that internet anonymity is reduced, suggesting that individuals face more public situations. Individuals tend to engage in behaviors that match the environment to maintain the impression of their real identity. What kind of behavior matches the environment comes from the observation about audiences' responses⁵⁸. In other words, when people observe that cyberbullying is unwanted, they will actively avoid such behaviors in this case. After people's motivation for impression management is activated, they select what kind of behavior is appropriate to engage in, which is usually purposeful⁵⁰. In the context of this study, cyberbullying is restrained for personal social and reputation maintenance.

Fourthly, this study expands the research of impression management theory in the digital context. In the past, this theory was mostly used to explain people's face-to-face impression management behaviors in organizations, such as interviews^{105–107} and other employee behaviors within organizations^{87–89,108–111}. However, with societal changes and technological advancements, the network has gradually become an important channel for people's social activities. The network creates a new environment that is different from face-to-face situations. In particular, digital interaction often means a higher degree of anonymity, and individuals are more likely to engage in negative behaviors under the protection of virtual images¹⁰. The IP location display provides a way to reduce the degree of internet anonymity, making the virtual image of the individual closer to its real image to a certain extent, thereby activating the individual's impression management motivation to inhibit cyberbullying.

Implications for practice

The research results provide practical enlightenment for policy makers, social platform enterprises, and other stakeholders potentially involved in cyberbullying.

For policy makers, as the popularity of the internet has gradually increased worldwide, cyberbullying has emerged as a challenging social governance issue. Consequently, the Chinese government has proposed a mandatory display of users' IP location information. Given that there is no precedent for this in other countries or historical contexts, this initiative is both innovative and experimental, despite its effectiveness not being confirmed beforehand. This study empirically validates the effectiveness of IP location display in suppressing cyberbullying and elucidates its mechanism. This provides theoretical support for policy makers to continue implementing the policy and offers insights for the subsequent refinement of relevant regulations and legal provisions. Since the suppression of cyberbullying through IP location display underscores the role of reduced internet anonymity in controlling cyberbullying, policy makers may further consider requiring social media opinion leaders to adhere to stricter disclosure rules for real identity information based on this rationale. This is because opinion leaders wield significant social influence and, if they engage in cyberbullying, their followers may be more prone to follow suit, thereby necessitating greater responsibilities and constraints. Additionally, it inspires policy makers to explore more solutions that can enhance individuals' motivation to engage in impression management in the virtual world. For instance, individuals who exhibit uncivilized behavior online could be prohibited from posting on public channels for a specified duration and required to publish an apology to the victim on WeChat Moments. Compared to other platforms like Weibo, Douyin, and Bilibili, WeChat Moments is a more private and intimate online social channel that is more closely tied to individuals' real identities, potentially fostering greater impression management motivation.

For social platform enterprises, they serve as frontline executors in complying with relevant policies and combating cyberbullying. Their actions directly impact the effectiveness of policies aimed at curbing cyberbullying, and addressing cyberbullying also constitutes a part of their corporate social responsibility⁴⁸. Hence, based on the conclusion that IP location display effectively inhibits cyberbullying, social platform enterprises must strictly enforce the policy of mandatory IP location information display and further enhance its visibility, enabling individuals to more readily see their own and others' IP location information. Moreover, from an impression management perspective, IP location display operates as a punishment-like mechanism that deters individuals from engaging in cyberbullying to minimize impression loss. However, impression management can also serve as an incentive mechanism for individuals. For example, when a user consistently refrains from posting uncivilized words and deeds, the social platform enterprise can reward them with a medal of honor and a free one-week membership as a gesture of appreciation. These incentives will be broadcast across the platform and the user will be specially highlighted. For individuals who receive this honor, they will be more inclined to refrain from cyberbullying in the future to maintain their positive online image. Similarly, other bystander users may better regulate their behavior in response to this positive social reinforcement.

For all online users, this study introduces group identity as a boundary condition influencing the effect of IP location display on cyberbullying through impression management motivation. The results indicate that group identity plays a positive role, suggesting that people's behavior can be regulated by group norms. Social networking platforms host numerous groups formed by netizens sharing common interests, such as fan clubs, or extensions of formal organizations, like the Xiaomi user community. Group members with higher social status and reputation typically exert greater influence on other group members^{112,113}. They bear a greater responsibility to regulate their own and other group members' behavior. On the one hand, they need to set an example by refraining from cyberbullying to establish a positive image; on the other hand, if cyberbullying occurs within the group, they should remain neutral, mediate and mitigate conflicts, and control the spread of negative effects. Furthermore, formal and clear group norms need to be established within the group to regulate members' behavior. For instance, in some private communications, group administrators will post announcements requiring members to communicate civilly, with non-compliance resulting in exclusion from the group.

Limitations and future research directions

Although this study offers several theoretical and practical insights, it also has certain limitations that highlight avenues for future investigation. Firstly, although China's mandatory IP-location disclosure policy makes Chinese social media a unique research context for this study, the Chinese cultural and regulatory context may limit the generalizability of our findings. Chinese social media environments feature distinct social norms, including attitudes toward privacy, conformity, and group behavior, and stricter government regulations compared to more open Western digital environments. As a result, the effectiveness and psychological responses to IP location disclosure observed here may not directly translate to platforms operating under different cultural values or regulatory regimes. Future research should examine whether similar anti cyberbullying effects hold in more liberal or Western contexts, where user privacy norms and the impact of identity transparency might differ.

Secondly, we operationalized group identity primarily through users' IP locations because regional affiliation is a salient identity cue that users readily recognize in the Chinese social media context. However, it captures only one dimension of an individual's group identity. In reality, people simultaneously belong to multiple social groups defined not only by geography but also by factors such as ideology, interests, or demographics. Focusing exclusively on geographic identity may overlook these other important identity cues. Future research should incorporate multiple identity indicators, such as ideological affiliations or interest based communities, to determine whether the moderating effect of group identity on impression management and cyberbullying holds across different forms of identity.

Thirdly, this study employs self-reported measures of cyberbullying in an experimental setting. Because cyberbullying is a socially undesirable behavior, participants might under-report their involvement to meet social expectations, potentially introducing bias into the results. We attempted to minimize this issue by adopting techniques to reduce social desirability bias (e.g., ensuring anonymity, encouraging honest responses, and using tactful questioning) as suggested by prior studies^{1,97}. These precautions helped keep any such bias within a reasonable range, as evidenced by the significant differences observed between the IP-display and no-display conditions. However, due to platform permission limitations, we were unable to obtain real-world behavioral data (e.g., actual commenting records) from before and after the IP disclosure policy, which precluded us from performing a difference-in-differences analysis to directly evaluate the policy's impact. Therefore, we encourage future research to cross-validate our findings with observational data. For instance, researchers could compare the volume of reported abusive comments on social media platforms before versus after the implementation of IP location display to provide more objective evidence of this intervention's effectiveness.

Fourthly, although we conducted three independent experiments across different major social platforms (Douyin, Bilibili, and Weibo) to enhance the robustness and external validity of our findings, this approach has inherent limitations. The experimental scenarios were controlled and the data collection period was relatively short; hence, the results may not fully capture users' natural behavior over extended periods or in real-world settings. It remains uncertain whether the inhibitory effect of IP location disclosure on cyberbullying would persist (or possibly diminish) in an organic environment over the long run, and whether any long-term unintended consequences might emerge. Future studies should consider longitudinal or field designs that track actual user interactions over time. Such approaches would help determine if the observed effects of IP disclosure are transient or sustained, thereby improving the ecological validity of the findings.

Fifthly, the results indicate that impression management motivation only partially mediates the impact of IP location display on cyberbullying. In response to this, we introduced two alternative explanatory mechanisms in Experiment 3 (moral disengagement and privacy concerns) and found that these also had significant but partial mediating effects. This suggests that IP disclosure may influence cyberbullying through additional psychological processes beyond those examined in our study. Similarly, while we found that group identity (as operationalized in this study) significantly moderated the effect via impression management motivation, other individual or contextual factors that we did not capture. For example, personality traits (such as an individual's gaming orientation or level of neuroticism) might also shape this relationship. Future research could explore such personal characteristics or environmental factors as additional moderators to expand the theoretical framework and boundary conditions of the IP disclosure effect.

Finally, while displaying IP locations appears to deter cyberbullying, it also raises concerns about user privacy and potential adverse impacts on certain users. In Experiment 3, we observed that IP location display not only significantly increased users' impression management motivation and decreased moral disengagement, but also led to a rise in privacy concerns. Notably, this increase in privacy concerns did not translate into a further reduction in cyberbullying, implying that users who felt uneasy about IP disclosure were likely already cautious about their online behavior. In other words, regardless of the IP display, those individuals would not engage in cyberbullying, but the presence of IP tagging made them more anxious about their personal information being exposed. In particular, some vulnerable or marginalized users might not experience any boost in impression management motivation from IP transparency; rather, they could feel threatened or stressed by being visibly identified, a reaction akin to privacy fatigue. Such anxiety or avoidance in response to mandatory location disclosure could decrease these users' willingness to participate in online discussions, increasing the psychological cost of engaging on the platform. This unintended consequence may be detrimental to the harmonious development of online communities. Therefore, future research should investigate the delicate balance between deterring cyberbullying through transparency and safeguarding users' privacy and comfort, in order to find an optimal equilibrium between these objectives.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to participant privacy considerations, but are available from the corresponding author on reasonable request.

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Author contributions

Xuesheng Meng contributed to the conception and design of the study, collected and processed the data, conducted the statistical analyses, developed the theoretical framework, and wrote and revised the main manuscript text. Xu Cai contributed to the design of the study, the theoretical framework, the revision of the manuscript, and language editing. Jiaoyang Li contributed to the conception and design of the study, the theoretical framework, the revision of the manuscript, and led funding acquisition and project supervision. All authors reviewed and approved the final version.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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